

PE-EC801B

Fiber Optic Communication

Combined Module Notes

Modules 1 – 5

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Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Module 1

Vector Nature of Light — EM Wave Fundamentals

1. Vector Nature of Light — EM Wave Fundamentals

Light as an Electromagnetic Wave

Light is a **transverse electromagnetic (TEM) wave** in which the electric field $\vec{E}$ and magnetic field $\vec{H}$ are perpendicular to each other and to the direction of propagation $\hat{k}$. In free space, both fields satisfy Maxwell's wave equation.

1.1 Maxwell's Equations and Wave Equation

For a linear, isotropic, source-free dielectric medium (permittivity ϵ , permeability μ), Maxwell's curl equations are:

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}, \quad \nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

Taking the curl of the first equation and using the second gives the vector wave equation:

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

The speed of light in the medium is $v = 1/\sqrt{\mu\epsilon}$. In free space, $v = c = 3 \times 10^8$ m/s, where $c = 1/\sqrt{\mu_0\epsilon_0}$.

1.2 E and H Field Relationship

For a plane wave propagating in the $+z$ direction with wavenumber $k = \omega\sqrt{\mu\epsilon}$:

$$\vec{E} = E_0 e^{j(\omega t - kz)} \hat{x}, \quad \vec{H} = \frac{E_0}{\eta} e^{j(\omega t - kz)} \hat{y}$$

where $\eta = \sqrt{\mu/\epsilon}$ is the intrinsic impedance of the medium. In free space, $\eta_0 = 377 \Omega$.

Key EM Wave Parameters

Wave number	$k = \frac{2\pi}{\lambda} = \frac{\omega}{v} = \omega\sqrt{\mu\epsilon}$
Refractive index	$n = \frac{c}{v} = \sqrt{\mu_r\epsilon_r} \approx \sqrt{\epsilon_r}$ (non-magnetic)
Intrinsic impedance	$\eta = \sqrt{\mu/\epsilon}$; free space $\eta_0 = 377 \Omega$
Poynting vector	$\vec{S} = \vec{E} \times \vec{H}$ (power flow direction)

2. Propagation of Light — Reflection, Refraction, and TIR

Snell's Law and Critical Angle

When light travels from medium 1 (refractive index n_1) to medium 2 (n_2), **Snell's law** governs the angle of refraction: $n_1 \sin \theta_1 = n_2 \sin \theta_2$. The **critical angle** θ_c is the angle of incidence in medium 1 at which the refracted ray in medium 2 travels along the interface ($\theta_2 = 90^\circ$). **Total Internal Reflection (TIR)** occurs for $\theta_1 > \theta_c$ when $n_1 > n_2$.

Snell's Law and Critical Angle Formulas

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (\text{Snell's Law})$$

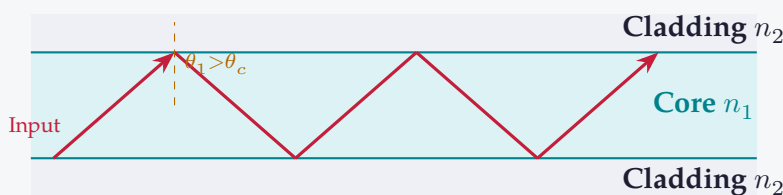
$$\sin \theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2) \Rightarrow \theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

TIR condition: $\theta_1 \geq \theta_c$ (light stays in the denser medium).

2.1 Propagation in a Cylindrical Dielectric Rod (Optical Fiber)

An optical fiber is a cylindrical dielectric rod comprising a **core** (refractive index n_1) surrounded by a **cladding** (refractive index $n_2 < n_1$). Light injected at the core-cladding interface at an angle $\geq \theta_c$ undergoes TIR repeatedly and is guided along the fiber without escaping.

TIR in an Optical Fiber — Ray Path



3. Ray Model — Acceptance Angle and Numerical Aperture

Numerical Aperture (NA) and Acceptance Angle

The **acceptance angle** θ_a (or half-cone angle) is the maximum angle at which light may enter the fiber end-face and still undergo TIR. The **Numerical Aperture (NA)** is a dimensionless measure of the light-gathering ability of the fiber: $NA = n_0 \sin \theta_a$ where n_0 is the refractive index of the medium at the input (usually air, $n_0 = 1$).

3.1 Derivation of NA

Consider a ray entering the flat end of the fiber at angle θ_i (in air). By Snell's law at the end-face:

$$\sin \theta_i = n_1 \sin \theta_r$$

where θ_r is the refraction angle inside the core. For TIR at the core-cladding interface, the ray must strike that interface at an angle $\geq \theta_c$, meaning the complementary angle $\theta_r \leq (90^\circ - \theta_c)$. Using $\sin \theta_c = n_2/n_1$:

Numerical Aperture — Complete Formula

$$NA = \sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$$NA = n_1 \sqrt{2\Delta}, \quad \Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \quad (\text{relative refractive index difference})$$

Acceptance angle: $\theta_a = \sin^{-1}(NA)$ (full acceptance cone = $2\theta_a$).

3.2 Meridional vs Skew Rays

- **Meridional rays:** Rays that pass through the fiber axis. Their path lies entirely in a plane containing the axis. They follow simple zigzag trajectories and are the easiest to analyze.
- **Skew rays:** Rays that do not pass through the fiber axis. They travel in a helical path around the axis. They have a larger acceptance angle than meridional rays and carry higher-order modes.

Worked Example: Calculate NA and Acceptance Angle

Given: $n_1 = 1.48$, $n_2 = 1.46$. Find NA and θ_a .

$$NA = \sqrt{1.48^2 - 1.46^2} = \sqrt{2.1904 - 2.1316} = \sqrt{0.0588} \approx 0.2425$$

$$\theta_a = \sin^{-1}(0.2425) \approx 14.03^\circ, \quad \Delta = \frac{1.48 - 1.46}{1.48} \approx 1.35\%$$

4. Wave Model — Modes and V-Number

Modes in an Optical Fiber

A **mode** is a distinct spatial distribution (pattern) of the electromagnetic field that can propagate along the fiber without changing shape. Only certain discrete field patterns satisfy the boundary conditions at the core-cladding interface; these are the allowed or guided modes.

4.1 Normalized Frequency (V-Number)

The **V-number** (also called the normalized frequency or normalized waveguide parameter) determines how many modes a fiber can support.

V-Number Formula

$$V = \frac{2\pi a}{\lambda} \text{NA} = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

where a = core radius, λ = free-space wavelength of light.

- $V < 2.405 \Rightarrow$ **single-mode** fiber (only the HE_{11} fundamental mode propagates).
- $V \geq 2.405 \Rightarrow$ **multimode** fiber.
- Total number of modes (step-index): $M \approx \frac{V^2}{2}$.
- Total number of modes (graded-index, parabolic): $M \approx \frac{V^2}{4}$.

4.2 Cutoff Condition and Mode Designation

Modes are designated HE_{mn} or EH_{mn} (hybrid modes), TE_{0n} , TM_{0n} . The fundamental mode HE_{11} has no cutoff. The first higher-order mode TE_{01} cuts off at $V = 2.405$ (first zero of Bessel function J_0).

4.3 Modal Analysis of a Step-Index Fiber

The electromagnetic field inside the core satisfies the wave equation in cylindrical coordinates, yielding Bessel functions J_n (oscillatory, inside core) and modified Bessel functions K_n (decaying, inside cladding). Applying continuity of tangential E and H at the core-cladding boundary gives the **characteristic equation** (eigenvalue equation) for the allowed propagation constants β :

Characteristic Equation for Step-Index Fiber

$$\left[\frac{J'_n(U)}{U J_n(U)} + \frac{K'_n(W)}{W K_n(W)} \right] \left[\frac{n_1^2 J'_n(U)}{U J_n(U)} + \frac{n_2^2 K'_n(W)}{W K_n(W)} \right] = n^2 \left(\frac{1}{U^2} + \frac{1}{W^2} \right)^2$$

where the normalized transverse wave numbers are:

$$U = a \sqrt{n_1^2 k_0^2 - \beta^2}, \quad W = a \sqrt{\beta^2 - n_2^2 k_0^2}, \quad U^2 + W^2 = V^2$$

For the LP_{01} (HE_{11}) **fundamental mode**: the simplified characteristic equation is $J_0(U)K_1(W) + J_1(U)K_0(W) = 0$. This transcendental equation is solved numerically to find β for each mode.

Key results from modal analysis:

- Each solution β_m corresponds to one guided mode with phase velocity $v_{ph} = \omega/\beta_m$.
- Group velocity: $v_g = d\omega/d\beta$ (determines pulse propagation and intermodal dispersion).
- For LP modes: LP_{lm} modes correspond to $HE_{l+1,m}$ and $EH_{l-1,m}$ modes. The LP approximation ($n_1 \approx n_2$) simplifies the characteristic equation to $J_{l-1}(U)/[U J_l(U)] = -K_{l-1}(W)/[W K_l(W)]$.

Worked Example: V-Number and Mode Count

Given: $a = 25 \mu\text{m}$, $\lambda = 1300 \text{ nm}$, $n_1 = 1.48$, $n_2 = 1.46$.

NA = 0.2425 (from previous example)

$$V = \frac{2\pi \times 25 \times 10^{-6}}{1300 \times 10^{-9}} \times 0.2425 = \frac{2\pi \times 25}{1.3} \times 0.2425 \approx 29.2$$

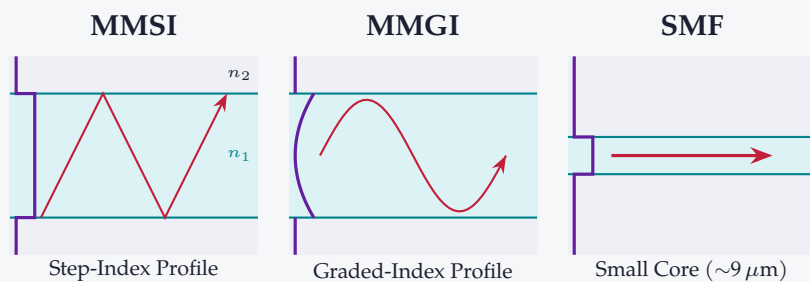
Number of modes $M \approx V^2/2 \approx 426$. Since $V \gg 2.405$, this is a multimode fiber.

5. Types of Optical Fibers

Classification of Optical Fibers

Optical fibers are classified by (a) **index profile** of the core — step-index (SI) or graded-index (GI), and (b) **number of modes** — single-mode (SM) or multimode (MM). The three main types are: Single-Mode Step-Index (SMF), Multimode Step-Index (MMSI), and Multimode Graded-Index (MMGI).

Refractive Index Profiles and Ray Paths — Three Fiber Types



5.1 Comparison Table — SMF vs MMSI vs MMGI

Parameter	SMF	MMSI	MMGI
Core diameter	$\sim 8\text{--}10\ \mu\text{m}$	$\sim 50\text{--}200\ \mu\text{m}$	$\sim 50\text{--}62.5\ \mu\text{m}$
Cladding diameter	$125\ \mu\text{m}$	$125\text{--}400\ \mu\text{m}$	$125\ \mu\text{m}$
Index profile	Step	Step	Parabolic ($\alpha \approx 2$)
V-number	< 2.405	$\gg 2.405$	$\gg 2.405$
No. of modes	1 (HE_{11} only)	$\sim V^2/2$ (hundreds)	$\sim V^2/4$
Bandwidth	Very high ($>10\ \text{GHz}\cdot\text{km}$)	Low ($\sim 10\text{--}25\ \text{MHz}\cdot\text{km}$)	Medium ($\sim 400\text{--}2000\ \text{MHz}\cdot\text{km}$)
Modal dispersion	None	High	Low (near-zero for $\alpha=2$)
NA	$\sim 0.1\text{--}0.12$	$\sim 0.2\text{--}0.5$	$\sim 0.2\text{--}0.3$
Coupling efficiency	Low (small core)	High	Medium
Light source	Laser only	LED or Laser	LED or Laser
Application	Long-haul, WDM	Short-haul, LAN	LAN, medium distance

Quick Revision — Module 1

Light Propagation and Optical Fiber Fundamentals | PE-EC801B

Key Formulas at a Glance

Snell's Law	$n_1 \sin \theta_1 = n_2 \sin \theta_2$
Critical angle	$\theta_c = \sin^{-1}(n_2/n_1) \quad (n_1 > n_2)$
Numerical Aperture	$NA = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta}$
Relative index diff.	$\Delta = (n_1 - n_2)/n_1$ (approx. for $n_1 \approx n_2$)
V-number	$V = (2\pi a/\lambda) NA$
SMF condition	$V < 2.405$
Mode count (MMSI)	$M \approx V^2/2$
Mode count (MMGI)	$M \approx V^2/4$
Wave speed in medium	$v = c/n; \quad n = \sqrt{\epsilon_r}$

One-Line Definitions

- **TIR:** Total Internal Reflection — light stays in denser medium when $\theta_i \geq \theta_c$.
- **NA:** Numerical Aperture — measure of light-gathering ability of the fiber.
- **V-number:** Normalized frequency parameter determining number of guided modes.
- **Meridional ray:** Ray passing through the fiber axis, zigzag path.
- **Skew ray:** Ray traveling in a helical path, does not pass through axis.
- **Mode:** Distinct stable field distribution pattern guided along the fiber.
- **SMF:** Single-mode fiber with $V < 2.405$; eliminates modal dispersion.
- **MMGI:** Multimode graded-index fiber; parabolic profile minimizes modal dispersion.

Frequently Asked / Viva Questions

Q1. What is Total Internal Reflection and when does it occur?

TIR occurs when light travels from a denser medium (n_1) to a rarer medium ($n_2 < n_1$) and the angle of incidence exceeds the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$. All incident light is reflected back; no energy enters medium 2.

Q2. Why is the refractive index of the core greater than the cladding in an optical fiber?

TIR — which is the guiding mechanism — requires light to travel from a denser

Frequently Asked / Viva Questions (contd.)

to rarer medium. Thus $n_1 > n_2$ is essential for guided propagation.

Q3. Define Numerical Aperture (NA). What does a higher NA imply?

$NA = \sqrt{n_1^2 - n_2^2}$ is the sine of the maximum acceptance angle. Higher NA means the fiber accepts light over a wider cone but also supports more modes (greater modal dispersion).

Q4. What is the V-number and what is its significance?

The V-number $= (2\pi a/\lambda) NA$ is the normalized frequency. It determines the number of guided modes. For $V < 2.405$, only one mode propagates (single-mode operation).

Q5. Distinguish between meridional and skew rays.

Meridional rays pass through the fiber axis and travel in a plane containing it. Skew rays do not intersect the axis; they travel helically. Skew rays have a slightly larger effective acceptance angle.

Q6. Why does a graded-index fiber have lower modal dispersion than step-index?

In MMGI, rays in the outer (faster) region travel longer paths but at higher speeds (lower n), while rays near the axis travel slower but shorter paths. This self-equalizing effect reduces pulse spreading.

Q7. What is the cutoff condition for single-mode operation?

$V < 2.405$, where 2.405 is the first zero of the Bessel function J_0 . Below this, only the fundamental HE_{11} mode propagates.

Q8. What is Δ (relative refractive index difference)?

$\Delta = (n_1 - n_2)/n_1 \approx (n_1^2 - n_2^2)/(2n_1^2)$. It is typically 0.1–2% for communication fibers. Larger Δ means higher NA but also higher dispersion.

Q9. How does the number of modes depend on V-number?

For step-index: $M \approx V^2/2$. For graded-index (parabolic): $M \approx V^2/4$. The graded profile halves the mode count for the same V-number.

Q10. What are the dimensions of a standard SMF and MMSI fiber?

SMF: core $\sim 8\text{--}10\ \mu\text{m}$, cladding $125\ \mu\text{m}$. MMSI: core $50\text{--}200\ \mu\text{m}$, cladding $125\ \mu\text{m}$. MMGI: core $50\text{--}62.5\ \mu\text{m}$, cladding $125\ \mu\text{m}$.

Q11. State the wave equation for the electric field in a dielectric medium.

$\nabla^2 \vec{E} = \mu\epsilon \partial^2 \vec{E} / \partial t^2$. The speed of propagation is $v = 1/\sqrt{\mu\epsilon} = c/n$.

Q12. What is the Poynting vector and what does it represent in fiber optics?

$\vec{S} = \vec{E} \times \vec{H}$ (W/m^2). It represents the direction and magnitude of electromagnetic power flow. In a fiber, it points along the fiber axis for guided modes.

Summary: Fiber Type Comparison at a Glance

Feature	SMF	MMSI	MMGI
V-number	< 2.405	Large	Large
Core dia.	8–10 μm	50–200 μm	50–62.5 μm
Dispersion	Lowest	Highest	Low
Bandwidth	Highest	Lowest	Medium
Source	Laser	LED/Laser	LED/Laser
Use case	Long haul	Short links	LAN/campus

Module 2

Signal Attenuation in Optical Fibers

1. Signal Attenuation in Optical Fibers

Definition of Attenuation

Attenuation (or fiber loss) is the reduction in optical power as light propagates along a fiber. It is expressed in dB/km and is a primary factor limiting the repeater spacing in a fiber optic link.

1.1 Power Loss Formula

If P_{in} is the optical power launched into a fiber of length L and P_{out} is the power at the far end:

Attenuation Coefficient

$$\alpha \text{ (dB/km)} = \frac{10}{L} \log_{10} \left(\frac{P_{in}}{P_{out}} \right)$$

$$P_{out} = P_{in} \cdot 10^{-\alpha L/10}$$

At 1310 nm: $\alpha \approx 0.35$ dB/km; at 1550 nm: $\alpha \approx 0.20$ dB/km (for standard SMF).

1.2 Causes of Attenuation

1. Absorption Losses:

- **Intrinsic absorption:** Due to electronic resonances (UV) and vibrational resonances (IR) of the silica (SiO_2) matrix. This sets the fundamental lower limit on attenuation.
- **Extrinsic absorption:** Caused by OH^- ions (hydroxyl impurities from water vapor during fabrication) — produces absorption peaks at 1.38 μm , 950 nm, and 725 nm. Also caused by metal ion impurities (Fe, Cu, Mn).

2. Scattering Losses:

- **Rayleigh scattering:** Caused by random microscopic density and composition fluctuations (dimensions $\ll \lambda$) frozen in during fiber fabrication. Varies as λ^{-4} — dominant at short wavelengths. The Rayleigh scattering coefficient $A_R \approx 0.8$ dB/km $\cdot \mu\text{m}^4$.
- **Mie scattering:** Due to inhomogeneities comparable in size to the wavelength (imperfect core-cladding interfaces, diameter fluctuations, bubbles). Usually small in modern fibers.

3. Bending Losses:

- **Macrobending:** When the fiber is bent to a radius $R < R_c$ (critical radius), the evanescent field at the outer edge must travel faster than the speed of light, so energy radiates out.
- **Microbending:** Caused by small random deformations of the fiber axis due to lateral stress during cabling. Causes mode coupling and energy transfer to radiation modes.

2. Dispersion in Optical Fibers

Dispersion — Definition

Dispersion is the spreading of optical pulses as they propagate along a fiber, caused by different frequency components or different modes traveling at different speeds. It limits the bandwidth and hence the data rate of the fiber link.

2.1 Types of Dispersion

Intermodal (Modal) Dispersion: Occurs in multimode fibers because different modes travel at different velocities. The fastest mode (along the axis) and the slowest mode differ in travel time. For MMSI:

$$\delta\tau_{\text{modal}} = \frac{n_1 L}{c} \cdot \Delta = \frac{n_1 L}{c} \cdot \frac{n_1 - n_2}{n_1}$$

Modal dispersion is **absent in single-mode fibers** and greatly reduced in MMGI fibers.

Intramodal (Chromatic) Dispersion: Occurs within a single mode and has two components:

1. **Material dispersion:** The refractive index $n(\lambda)$ varies with wavelength. Different spectral components of the source pulse travel at different speeds. Characterized by material dispersion parameter D_m (ps/nm·km). Zero at $\lambda_0 \approx 1.27 \mu\text{m}$ for silica.
2. **Waveguide dispersion:** Even for a fixed wavelength, the propagation constant β depends on the ratio $V = 2\pi a \text{NA}/\lambda$, so the group velocity varies with wavelength due to the fiber geometry. Significant in SMF. Characterized by D_w (ps/nm·km). Negative at 1550 nm.
3. **Chromatic (total) dispersion:** $D = D_m + D_w$ (ps/nm·km). For standard SMF: $D = 0$ at $\lambda \approx 1310 \text{ nm}$; $D \approx +17 \text{ ps/nm·km}$ at 1550 nm.

Dispersion Formulas

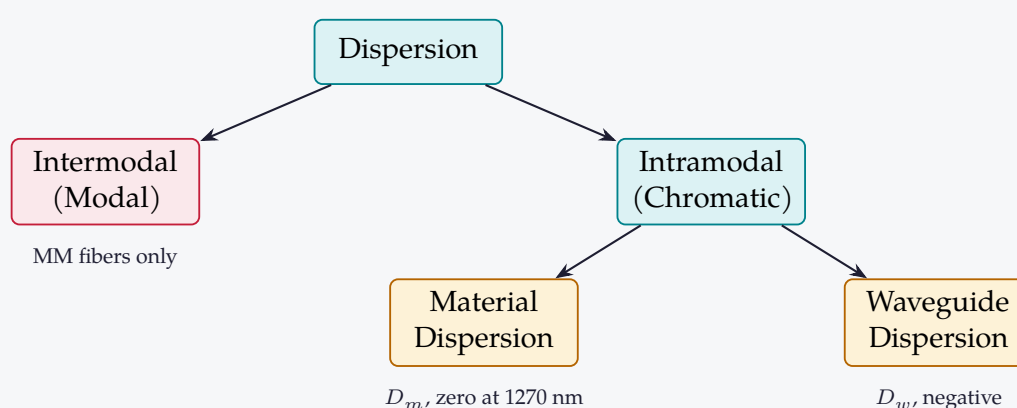
RMS pulse broadening (chromatic) $\sigma = D \cdot L \cdot \Delta\lambda$ (ps)

Bandwidth–length product $B \cdot L \leq \frac{0.44}{\sigma/L}$ (Gaussian pulse)

Modal dispersion (MMSI) $\delta\tau = \frac{n_1 L \Delta}{c}$

Modal dispersion (MMGI, $\alpha=2$) $\delta\tau \approx \frac{n_1 L \Delta^2}{2c}$

Dispersion Types — Classification Chart



3. Fiber Fabrication Processes

Overview of Fabrication Methods

Modern silica optical fibers are made by **Chemical Vapor Deposition (CVD)** processes, where dopant-laden gases react at high temperature to deposit glass layers. The four main processes are MCVD, OVD, VAD, and PCVD.

3.1 MCVD — Modified Chemical Vapor Deposition

Dopant gases (SiCl_4 , GeCl_4 , POCl_3 , BCl_3) are flowed inside a silica tube. An external oxygen–hydrogen flame traverses the tube, causing oxidation and deposition of glass soot inside. Multiple passes deposit core and cladding layers. The tube is then collapsed into a solid **preform** by heating to $> 2000^\circ\text{C}$.

3.2 OVD — Outside Vapor Deposition

Soot is deposited on the **outside** of a bait rod (mandrel) via a flame hydrolysis burner. The rod is removed and the porous soot boule is consolidated (sintered) in a furnace to form a clear glass preform, then drawn into fiber.

3.3 VAD — Vapor Phase Axial Deposition

Soot is deposited axially at the end of a rotating preform billet using burners. The porous preform grows **upward** and is continuously pulled into a consolidation furnace. Allows continuous fiber production. Produces very low OH^- content.

3.4 PCVD — Plasma Chemical Vapor Deposition

Uses a non-resonant microwave plasma (instead of a flame) inside a silica tube to drive the CVD reaction. Very precise compositional control; used for special fibers.

Feature	MCVD	OVD	VAD	PCVD
Deposition location	Inside tube	Outside rod	Axial end	Inside tube
Heat source	External flame	Flame hydrolysis	Flame	Microwave plasma
Collapse needed?	Yes	Yes (soot)	Yes	Yes
OH^- impurity	Moderate	Low	Very low	Low
Production rate	Medium	High	High (continuous)	Low
Profile control	Good	Good	Good	Excellent

4. Fiber Measurement Techniques

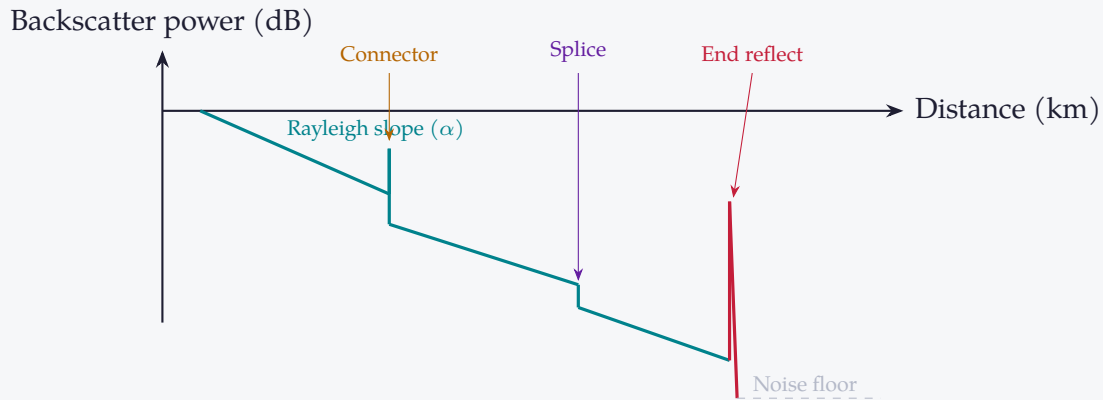
Optical Time-Domain Reflectometer (OTDR)

An OTDR is a sophisticated instrument that characterizes a fiber by injecting a short optical pulse and measuring the Rayleigh backscattered and Fresnel-reflected light as a function of time (and hence distance). It gives a **complete loss profile** of the fiber from a single end.

4.1 OTDR — Principle and Trace Analysis

The OTDR launches a pulse into the fiber. Light is continuously backscattered toward the OTDR. Splice losses, connector losses, and faults appear as steps or spikes. The time delay t between launch and detection corresponds to a distance $d = ct/(2n_1)$.

Typical OTDR Trace



Key OTDR terms:

- **Dead zone:** Distance near the launch connector where the receiver is saturated after a large Fresnel reflection — events within this zone cannot be detected. Typical: 1–10 m.
- **Dynamic range:** The difference between the initial backscatter power level and the noise floor (dB). Larger dynamic range $\Rightarrow$ longer fiber can be characterized.

4.2 Cutback Method — Attenuation Measurement

The most accurate method for measuring attenuation. Launch power P_1 is measured by cutting the fiber close to the input (~ 2 m from the source). Then the full-length measurement P_2 is taken. No need to disturb the launch conditions.

$$\alpha = \frac{10}{L} \log_{10} \left(\frac{P_1}{P_2} \right) \quad \text{dB/km}$$

4.3 Interferometric Method — Dispersion Measurement

Used to measure **group delay** vs. wavelength. A broadband or tunable source illuminates the fiber; the output is compared against a reference path in a Mach-Zehnder or Michelson interferometer. Phase shift vs. wavelength gives the group delay $\tau(\lambda)$, from which $D(\lambda) = (1/L) d\tau/d\lambda$ (ps/nm·km).

Quick Revision — Module 2

Signal Degradation, Fabrication, and Measurement | PE-EC801B

Key Formulas at a Glance

Attenuation coefficient	$\alpha = \frac{10}{L} \log_{10}(P_{in}/P_{out})$ (dB/km)
Output power	$P_{out} = P_{in} \cdot 10^{-\alpha L/10}$
Rayleigh scattering	Varies as λ^{-4}
RMS pulse spread	$\sigma = D \cdot L \cdot \Delta\lambda$
Modal dispersion (MMSI)	$\delta\tau = n_1 L \Delta/c$
Modal dispersion (MMGI)	$\delta\tau \approx n_1 L \Delta^2/(2c)$
Chromatic dispersion	$D = D_m + D_w$ (ps/nm·km)
BW-length product	$B \cdot L \leq 0.44/(\sigma/L)$
OTDR distance	$d = ct/(2n_1)$

One-Line Definitions

- **Attenuation:** Reduction in optical power along the fiber; expressed in dB/km.
- **Rayleigh scattering:** Scattering from density fluctuations; $\propto \lambda^{-4}$; dominant at short λ .
- **Modal dispersion:** Pulse broadening from different modes arriving at different times; absent in SMF.
- **Material dispersion:** Pulse broadening due to $n(\lambda)$ variation; zero at 1270 nm in silica.
- **Waveguide dispersion:** Pulse broadening due to wavelength-dependent propagation constant.
- **OTDR:** Instrument that measures fiber loss profile from a single end using backscattered light.
- **Dead zone:** Region near the OTDR launch where saturation prevents event detection.
- **Cutback method:** Gold-standard destructive technique for measuring fiber attenuation accurately.
- **MCVD:** Modified CVD — glass deposited inside a silica tube by an external traversing flame.

Frequently Asked / Viva Questions

Q1. What are the main causes of attenuation in an optical fiber?

Absorption (intrinsic UV/IR resonances and extrinsic OH^- impurities), Rayleigh scattering (λ^{-4}), Mie scattering, macrobending, and microbending losses.

Q2. Why is 1310 nm and 1550 nm preferred as operating windows?

At 1310 nm, chromatic dispersion is near zero for SMF. At 1550 nm, Rayleigh scattering is low, giving minimum attenuation (~ 0.2 dB/km) for standard silica SMF.

Q3. What is Rayleigh scattering and why is it important?

It results from random sub-wavelength density fluctuations in silica. Since it varies as λ^{-4} , it is the dominant loss at short wavelengths and sets the lower limit for SMF attenuation.

Q4. Distinguish between intermodal and intramodal dispersion.

Intermodal dispersion occurs in multimode fibers when different modes travel at different velocities. Intramodal (chromatic) dispersion occurs within a single mode due to material and waveguide effects.

Q5. Why does MMGI have lower modal dispersion than MMSI?

The parabolic index profile of MMGI makes higher-order modes travel faster (in lower-index outer regions), nearly equaling the speed of lower-order modes. This self-focusing effect reduces differential mode delay.

Q6. What is bandwidth–length product? Give a typical value.

It is the product $B \times L$ (MHz·km or GHz·km), measuring the information capacity of a fiber per unit length. Typical: MMSI ~ 20 MHz·km; MMGI ~ 1000 MHz·km; SMF > 100 GHz·km.

Q7. Explain the OTDR principle briefly.

A short optical pulse is launched into the fiber. The OTDR measures the time-resolved backscattered signal. Distance is calculated as $d = ct/(2n)$. The trace shows attenuation slope, splices, connectors, and fiber end.

Q8. What is the dead zone in an OTDR?

The dead zone is the distance near the OTDR launch connector where the receiver is saturated by the large Fresnel reflection, making it unable to detect nearby events. Typically 1–10 m.

Q9. Describe the cutback method for attenuation measurement.

Launch power P_1 is measured by cutting the fiber ~ 2 m from the source. Then full-length output P_2 is measured. $\alpha = (10/L) \log(P_1/P_2)$. Most accurate but destructive.

Q10. What is the difference between macrobending and microbending?

Macrobending is a large-radius bend that causes the evanescent field to radiate. Microbending consists of random small-amplitude axis deformations from cabling stress, causing mode coupling and loss.

Frequently Asked / Viva Questions (contd.)

Q11. What is MCVD? How is the preform made?

Modified CVD. Dopant gases flow inside a rotating silica tube; an external oxy-hydrogen flame deposits glass soot on the inner wall. Multiple passes build up core/cladding layers, then the tube is collapsed into a solid preform.

Comparison: Fabrication Methods

Method	Deposition	Heat Source	OH ⁻	Rate
MCVD	Inside tube	External flame	Moderate	Medium
OVD	Outside rod	Flame hydrolysis	Low	High
VAD	Axial end	Flame	Very low	High (cont.)
PCVD	Inside tube	Microwave plasma	Low	Low

Module 3

Optical Sources — LED

1. Optical Sources — LED

Light-Emitting Diode (LED)

An LED is a forward-biased p-n junction semiconductor device that emits light by **spontaneous emission** when electrons recombine with holes. LEDs for fiber optics operate in the 850 nm, 1310 nm, or 1550 nm windows.

1.1 Surface-Emitting vs Edge-Emitting LED

- **Surface-Emitting LED (SLED / Burrus LED):** Emits light perpendicular to the junction plane. Large emitting area ($\sim 50 \mu\text{m}$ diameter), Lambertian emission pattern, easy to couple to MMSI/MMGI fibers. Modulation bandwidth $\sim 20\text{--}100$ MHz.
- **Edge-Emitting LED (ELED):** Emits light parallel to the junction from the cleaved edge of the active stripe. Narrower emission cone, better coupling to single-mode or small-core multimode fibers. Modulation bandwidth $\sim 100\text{--}400$ MHz. Output spectrum FWHM $\sim 30\text{--}80$ nm.

LED Output and Modulation Bandwidth

$$P_{opt} \propto I_d \cdot \eta_{ext}, \quad P(\omega) = \frac{P(0)}{\sqrt{1 + (\omega\tau_r)^2}}$$

Modulation bandwidth: $f_{3dB} = \frac{\sqrt{3}}{2\pi\tau_r}$, where τ_r is the minority carrier radiative lifetime ($\sim 2\text{--}10$ ns).

Advantages of LEDs: Simple construction, lower cost, longer lifetime ($> 10^7$ hours), linear L-I characteristic, no threshold, operable without temperature control, immune to catastrophic failure.

Disadvantages: Broad output spectrum (large material dispersion), lower output power, slower modulation compared to laser diodes.

2. Optical Sources — Laser Diodes

Laser Diode

A **laser diode** (LD) is a semiconductor p-n junction in which **stimulated emission** dominates above a threshold current I_{th} . It produces coherent, narrow-spectrum, high-power output suitable for SMF and high-speed links.

2.1 Fabry-Perot (FP) Laser

Uses two cleaved facets as reflective mirrors forming a Fabry-Perot cavity. Oscillation occurs at wavelengths satisfying the resonance condition $L = m\lambda/(2n)$ (m integer). FP lasers are **multi-longitudinal-mode**; spectrum is a comb of modes separated by $\Delta\lambda = \lambda^2/(2nL)$.

2.2 DFB (Distributed Feedback) Laser

A Bragg grating is integrated into the active layer. Feedback is distributed along the cavity. Only the mode satisfying the Bragg condition $\lambda_B = 2n\Lambda$ (where Λ is the grating period) is amplified. DFB lasers are **single-longitudinal-mode**, essential for WDM systems.

2.3 Rate Equations and Threshold Current

Laser Rate Equations and L-I Characteristic

Carrier density N and photon density S :

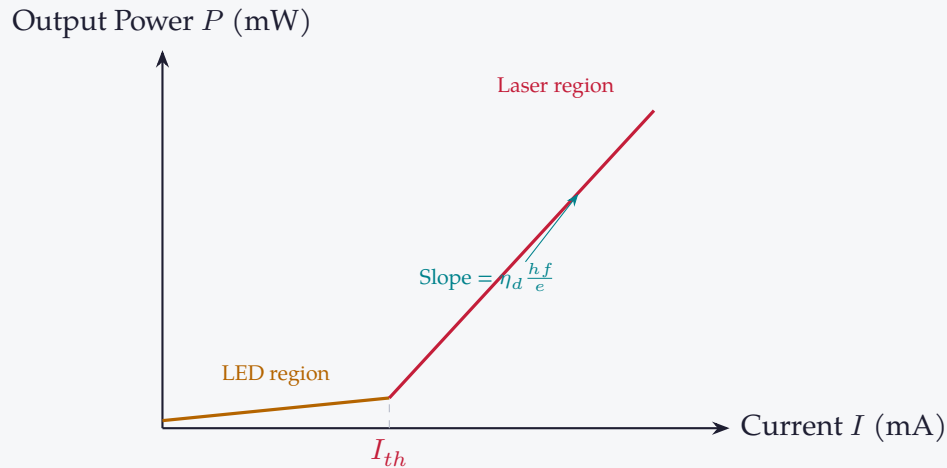
$$\frac{dN}{dt} = \frac{\eta_i I}{eV} - \frac{N}{\tau_n} - Gg_0(N - N_0)S$$

$$\frac{dS}{dt} = Gg_0(N - N_0)S + \frac{\beta N}{\tau_n} - \frac{S}{\tau_p}$$

Threshold current: I_{th} is where gain = loss (photon density starts growing exponentially).

$$P_{out} = \eta_d \cdot \frac{hf}{e}(I - I_{th}), \quad I > I_{th}$$

where η_d is differential quantum efficiency, hf is photon energy.

L-I Curve of a Laser Diode**3. Photodetectors — PIN Diode and APD****Photodetector — Responsivity and Quantum Efficiency**

A **photodetector** converts optical power into electrical current via the photoelectric effect. **Responsivity** $\mathcal{R}$ (A/W) is the photocurrent per unit optical power. **Quantum efficiency** η is the fraction of incident photons that generate electron-hole pairs.

Responsivity Formula

$$\mathcal{R} = \frac{I_p}{P_{opt}} = \frac{\eta e \lambda}{hc} \quad (\text{A/W})$$

where η = quantum efficiency, e = electron charge (1.6×10^{-19} C), λ = wavelength (m), h = Planck's constant (6.626×10^{-34} J·s), c = speed of light.
Also: $\mathcal{R} = \eta e / (hf)$ where $f = c/\lambda$ is optical frequency.

3.1 PIN Photodiode

A **PIN diode** has a lightly doped intrinsic (I) layer between heavily doped P and N regions. The wide depletion region extends across the I layer under reverse bias, allowing efficient absorption of photons over a large volume.

- **Structure:** $P^+ / I(\text{intrinsic}) / N^+$, reverse-biased.
- **Bandwidth:** Limited by transit time $\tau_{tr} = d/v_d$ and RC time constant. $f_{3dB} \approx 1/(2\pi\tau_{tr})$. Bandwidths up to tens of GHz are achievable.
- **Responsivity:** Typically $\mathcal{R} \approx 0.6\text{--}0.9$ A/W at 1300–1550 nm (InGaAs PIN).

3.2 Avalanche Photodiode (APD)

An APD operates under high reverse bias where photogenerated carriers gain sufficient energy to ionize more carriers through **impact ionization** — the avalanche effect. This

provides internal current gain M .

APD Gain, Noise, and SNR

APD multiplication gain: $M = I_{APD}/(I_{primary})$.

Excess noise factor: $F(M) \approx k_A M + (1 - k_A)(2 - 1/M)$ (McIntyre approximation), where k_A is the ionization ratio ($k_A = \alpha_h/\alpha_e$).

For InGaAs APD: M typically 10–30; $k_A \approx 0.4$ –0.7.

3.3 Noise in Photodetectors

- **Shot noise:** Due to the quantum nature of photon detection. $\langle i_s^2 \rangle = 2eI_p B$ (PIN); $\langle i_s^2 \rangle = 2eI_p M^2 F(M) B$ (APD), where B is bandwidth.
- **Thermal noise (Johnson noise):** Due to random thermal agitation of electrons in the load resistor R_L . $\langle i_T^2 \rangle = 4k_B T B / R_L$, where $k_B = 1.38 \times 10^{-23}$ J/K.
- **SNR:**

$$\text{SNR}_{PIN} = \frac{(\mathcal{R}P_{opt})^2}{2e\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

$$\text{SNR}_{APD} = \frac{(M\mathcal{R}P_{opt})^2}{2eM^2 F(M)\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

3.4 Optical Receivers — Pre-Amplifier Types

- **High-impedance (HZ) pre-amplifier:** Large R_L minimizes thermal noise, but limits bandwidth due to RC roll-off. Equalizer needed. Best sensitivity.
- **Transimpedance (TIA) pre-amplifier:** Uses feedback resistor R_f to convert photocurrent to voltage. Flat frequency response up to higher bandwidth with good sensitivity. Preferred in practice.

4. Optical Link Design — BER, Q-Factor, and Receiver Sensitivity

Optical link design is the process of ensuring that a fiber optic system meets both the **power budget** (sufficient received power for the target BER after accounting for all losses) and the **dispersion budget** (pulse broadening stays within the allowed bit-period fraction). The power budget gives: $P_{tx} - (\text{total losses}) \geq P_{rx,min}$, where losses include fiber attenuation (αL), splice losses, connector losses, and a system margin (~ 3 –6 dB). The dispersion budget requires: $\sigma_{total} \leq T_b/(4\sqrt{2})$ (Gaussian pulse limit), where $\sigma_{total} = \sqrt{\sigma_{modal}^2 + \sigma_{chromatic}^2 + \sigma_{source}^2}$.

BER and Q-Factor

Bit Error Rate (BER) is the probability that a received bit is in error. The **Q-factor** is the signal-to-noise amplitude ratio that directly maps to BER. For a Gaussian noise model: $\text{BER} = \frac{1}{2}\text{erfc}(Q/\sqrt{2})$.

Link Design Equations

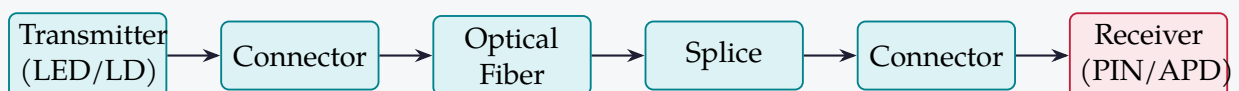
Q-factor	$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}$ (I_1, I_0 : mean currents for 1, 0)
BER vs Q	$BER \approx \frac{1}{2} \operatorname{erfc}\left(\frac{Q}{\sqrt{2}}\right) \approx \frac{\exp(-Q^2/2)}{Q\sqrt{2\pi}}$
BER = 10^{-9} requires	$Q \approx 6$
Quantum limit	$BER_{QL} = \frac{1}{2} e^{-\eta N_p}$, minimum $N_p \approx 10$ photons/bit
Receiver sensitivity	Minimum P_{min} (dBm) for target BER

4.1 Power Penalties

Power penalties are additional receiver power margins required due to system imperfections beyond the AWGN model.

Penalty Source	Cause	Typical Range
Modal noise	Mode partition noise in MMF; speckle pattern changes	0.5–1 dB
Extinction ratio	Non-zero optical power for “0” bit ($r = P_0/P_1 \neq 0$)	1–2 dB (for $r = 0.1$)
Timing jitter	Random deviation in bit timing $\Rightarrow$ ISI	0.5–2 dB
Chromatic dispersion	Pulse broadening causes ISI at high bit rate	Depends on $D \cdot B \cdot L$
Reflection noise	Multiple reflections create interferometric noise	0.5–1 dB

Block Diagram — Complete Fiber Optic Link



Attenuation + Dispersion

Quick Revision — Module 3

Optical Sources, Detectors, and Link Design | PE-EC801B

Key Formulas at a Glance

Responsivity	$\mathcal{R} = \eta e \lambda / (hc) \quad (\text{A/W})$
Shot noise (PIN)	$\langle i_s^2 \rangle = 2e\mathcal{R}P_{opt}B$
Thermal noise	$\langle i_T^2 \rangle = 4k_B T B / R_L$
LED modulation BW	$f_{3dB} = \sqrt{3} / (2\pi\tau_r)$
Laser output power	$P_{out} = \eta_d (hf/e)(I - I_{th}) \quad (I > I_{th})$
Q-factor	$Q = (I_1 - I_0) / (\sigma_1 + \sigma_0)$
BER (Gaussian)	$\text{BER} = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$
BER = 10^{-9}	$Q \approx 6$
APD SNR	$\propto M^2 / F(M) \quad (\text{optimum } M \text{ exists})$

One-Line Definitions

- **Spontaneous emission:** Random photon emission by an excited electron without external stimulus (LED).
- **Stimulated emission:** A passing photon triggers an identical photon from an excited state (Laser).
- **Threshold current I_{th} :** The bias current above which laser gain exceeds cavity loss.
- **DFB laser:** Distributed Feedback laser; single-mode due to integrated Bragg grating.
- **Responsivity $\mathcal{R}$:** Photocurrent generated per watt of incident optical power (A/W).
- **Quantum efficiency η :** Fraction of photons that create an electron-hole pair.
- **APD gain M :** Internal multiplication of photocurrent by impact ionization.
- **Q-factor:** SNR amplitude ratio; $Q = 6$ gives $\text{BER} = 10^{-9}$.
- **Transimpedance amp:** Pre-amplifier with feedback R_f giving flat bandwidth and good sensitivity.

Frequently Asked / Viva Questions

Q1. What is the difference between an LED and a Laser diode?

LED uses spontaneous emission; broad spectrum; no threshold; lower power. Laser diode uses stimulated emission above I_{th} ; narrow coherent spectrum; higher power; higher modulation speed.

Frequently Asked / Viva Questions (contd.)

Q2. What is the threshold current in a laser diode?

It is the minimum bias current at which the gain of the active medium equals the total cavity losses, causing the onset of lasing (stimulated emission dominates).

Q3. Why is a DFB laser preferred for WDM systems?

A DFB laser emits a single longitudinal mode (due to the Bragg grating), giving a very narrow linewidth (<1 MHz). This is essential to prevent channel overlap in dense WDM systems.

Q4. Define responsivity and derive the formula $\mathcal{R} = \eta e \lambda / hc$.

$\mathcal{R} = I_p / P_{opt}$. Each photon of energy $hf = hc/\lambda$ generates η electron-hole pairs, contributing charge ηe . Per second, $P_{opt}/(hf)$ photons arrive, giving $I_p = \eta e P_{opt}/(hf) = \eta e \lambda P_{opt}/(hc)$, so $\mathcal{R} = \eta e \lambda / (hc)$.

Q5. What is the advantage of APD over PIN diode?

APD provides internal current gain M through avalanche multiplication, improving receiver sensitivity (lower minimum detectable power). However, it introduces excess noise $F(M)$ and requires higher bias voltage.

Q6. What is shot noise? When is it dominant?

Shot noise arises from the discrete quantum nature of photon detection. $\langle i_s^2 \rangle = 2eI_p B$. It dominates over thermal noise at high optical power or when the load resistor is large.

Q7. Define Q-factor and state its relation to BER.

$Q = (I_1 - I_0)/(\sigma_1 + \sigma_0)$. For Gaussian noise: $\text{BER} = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$. $\text{BER} = 10^{-9}$ requires $Q \approx 6$.

Q8. What is the quantum limit of detection?

The absolute minimum BER is set by the Poisson statistics of photon arrival: $\text{BER} = \frac{1}{2} e^{-\eta N_p}$. The quantum limit is ~ 10 photons/bit for ideal detection.

Q9. What is extinction ratio penalty?

When the laser emits non-zero power for a "0" bit (extinction ratio $r = P_0/P_1 \neq 0$), the eye opening reduces, requiring extra received power to maintain the same BER. $\text{Penalty} = 10 \log_{10} \left(\frac{1+r}{1-r} \right)$.

Q10. Compare surface-emitting and edge-emitting LEDs.

SLED: wide emitting area, Lambertian pattern, easy coupling to MMF, BW ~ 100 MHz. ELED: narrow emission cone, better coupling to SMF or small MMF, higher BW ~ 400 MHz, higher power.

Q11. What is the difference between high-impedance and transimpedance pre-amplifiers?

HZ: large R_L minimizes thermal noise; limited bandwidth needing equalization. TIA: uses feedback R_f ; flat bandwidth; moderate noise; preferred for practical high-speed receivers.

LED vs Laser Diode — Comparison

Parameter	LED	Laser Diode
Emission type	Spontaneous	Stimulated
Coherence	Incoherent	Coherent
Spectral width	30–80 nm	<1 nm (DFB)
Threshold	None	I_{th} required
Modulation BW	~100–400 MHz	>10 GHz
Output power	Low (mW)	High (>10 mW)
Temperature sensitivity	Low	High
Cost	Low	High
Application	Short-haul, MMF	Long-haul, SMF, WDM

Module 4

Optical Switches

1. Optical Switches

Optical Switch

An **optical switch** is a device that selectively routes an optical signal from one port to another without converting it to an electrical signal. Switches are essential for optical cross-connects (OXC), protection/restoration, and reconfigurable networks.

1.1 Classification of Optical Switches

Type	Mechanism	Switching Time	Remarks
Mechanical	Physical fiber/mirror movement	ms to s	Low loss; no polarization dep.; slow
Electro-optic	Pockels effect in LiNbO_3	ps to ns	Fast; polarization dependent; insertion loss
Acousto-optic	Acousto-optic diffraction (Bragg)	μs	Can switch wavelengths; tunable
Thermo-optic	Thermo-optic effect in waveguide	ms	Low cost (Si/polymer); compact
MEMS	Micro-mirror arrays (MEMs)	ms	Large port count; low loss

2. Directional Couplers — Coupled Mode Analysis

Directional Coupler

A **directional coupler** is a 4-port device formed by bringing two waveguides or fibers into close proximity so that their evanescent fields overlap. Power is transferred from one waveguide to the other via evanescent coupling over the coupling length.

2.1 Coupled Mode Equations

Let $a_1(z)$ and $a_2(z)$ be the field amplitudes in waveguides 1 and 2. Coupled mode theory gives:

$$\frac{da_1}{dz} = -j\kappa a_2 e^{j\Delta\beta z}, \quad \frac{da_2}{dz} = -j\kappa a_1 e^{-j\Delta\beta z}$$

where κ is the coupling coefficient (dependent on overlap integral of evanescent fields) and $\Delta\beta = \beta_1 - \beta_2$ is the phase mismatch.

Power Transfer in a Directional Coupler

For identical waveguides ($\Delta\beta = 0$, perfect phase matching), the power in each arm as a function of length L :

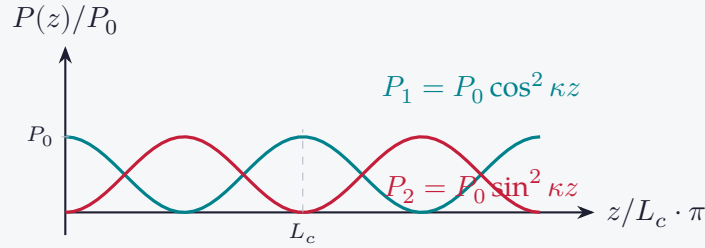
$$P_1(L) = P_0 \cos^2(\kappa L), \quad P_2(L) = P_0 \sin^2(\kappa L)$$

Coupling length: $L_c = \pi/(2\kappa)$ — full power transfer from waveguide 1 to 2.

For partial coupling ratio ϕ : $L = \phi\pi/(2\kappa)$.

With phase mismatch: $\max(P_2/P_0) = \kappa^2/(\kappa^2 + (\Delta\beta/2)^2)$.

Directional Coupler — Power Transfer Along Coupling Length



3. Mach-Zehnder Modulator (MZM)

Mach-Zehnder Modulator (MZM)

A **Mach-Zehnder Modulator** is an electro-optic intensity modulator based on LiNbO_3 . Light is split into two arms of a Mach-Zehnder interferometer; an applied electric field via the Pockels effect changes the refractive index of one (or both) arms, causing a phase shift. When the two beams recombine, constructive or destructive interference produces intensity modulation.

3.1 Structure and Operation

1. Input light is split 50:50 at a Y-junction.
2. Electrodes on each arm apply voltage V across the waveguide; the Pockels effect changes the refractive index $\Delta n \propto r_{33}E$ where r_{33} is the electro-optic coefficient ($\approx 30 \text{ pm/V}$ for LiNbO_3).
3. A phase shift $\Delta\phi = \pi V/V_\pi$ is induced.

4. The two arms recombine at a second Y-junction.

MZM Transfer Function

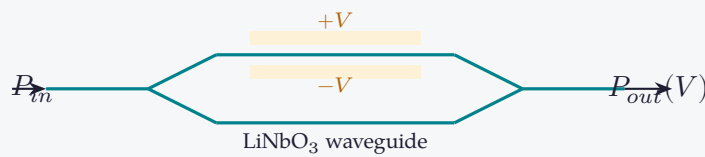
$$P_{out}(V) = P_{in} \cos^2\left(\frac{\pi V}{2V_\pi}\right) = \frac{P_{in}}{2} \left[1 + \cos\left(\frac{\pi V}{V_\pi}\right) \right]$$

Half-wave voltage V_π : The voltage required to shift the phase by π (switches from max to min transmission).

$$\text{Extinction ratio (ER): } r_{ext} = P_{max}/P_{min} \quad (\text{dB}) = 10 \log_{10}(r_{ext})$$

Ideal MZM: $r_{ext} \rightarrow \infty$ (perfect extinction). Practical: 20–30 dB.

Mach-Zehnder Modulator — Structure and Output



4. EDFA — Erbium-Doped Fiber Amplifier

EDFA

An **Erbium-Doped Fiber Amplifier (EDFA)** is an optical amplifier that amplifies light directly in the optical domain by stimulated emission from excited Er^{3+} ions. It operates in the 1530–1565 nm C-band (or 1570–1610 nm L-band), exactly matching the lowest-loss window of silica fiber.

4.1 Structure and Pumping

An EDFA consists of a few meters of silica fiber doped with Er^{3+} ions. A pump laser excites the erbium ions to higher energy levels. Two standard pump wavelengths:

- **980 nm pump:** Three-level pumping; low noise figure ($\sim 3\text{--}4$ dB); requires WDM coupler.
- **1480 nm pump:** Two-level pumping (directly pumps to metastable $^4I_{13/2}$ level); higher output power; noise figure $\sim 4\text{--}5$ dB.

4.2 Gain Spectrum and ASE Noise

EDFA provides gain over ~ 35 nm bandwidth (1530–1565 nm). The gain spectrum is not flat; tilt compensation requires gain flattening filters (GFF).

Amplified Spontaneous Emission (ASE): Not all spontaneous photons escape the fiber; a fraction is guided along the fiber and amplified, producing broadband noise. The ASE power spectral density in each polarization is:

$$S_{ASE}(\nu) = n_{sp} h\nu (G - 1)$$

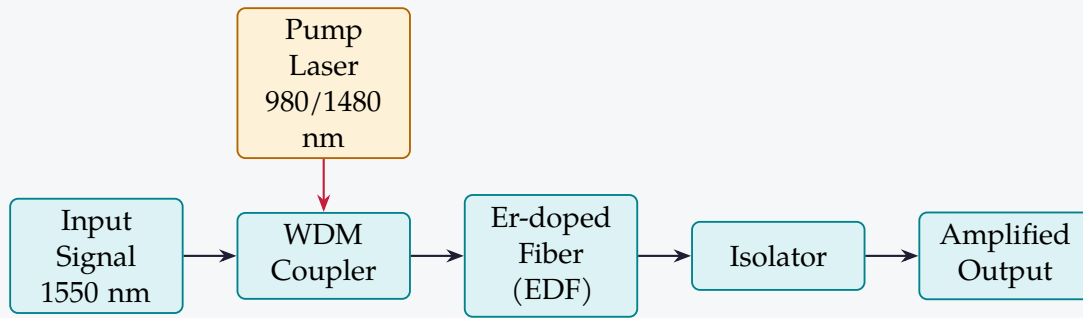
where $n_{sp} \geq 1$ is the spontaneous emission factor ($n_{sp} = N_2/(N_2 - N_1)$ for a two-level system).

Noise Figure (NF) of an EDFA:

$$NF = 2n_{sp} \frac{G-1}{G} + \frac{1}{G} \approx 2n_{sp} \quad \text{for large } G$$

Minimum theoretical NF = 3 dB (for $n_{sp} = 1$, ideal inversion).

EDFA Structure — Forward Pumped Configuration



5. Raman Amplifier

Raman Amplifier

A **Raman amplifier** exploits **Stimulated Raman Scattering (SRS)** in silica fiber to amplify an optical signal. A high-power pump transfers energy to a signal that is Stokes-shifted by ~ 13 THz (corresponding to ~ 100 nm red shift). The transmission fiber itself acts as the gain medium.

5.1 Stimulated Raman Scattering (SRS)

A pump photon at frequency ν_p interacts with a silica molecular vibration to produce a signal photon at lower frequency $\nu_s = \nu_p - \nu_v$ (Stokes shift) and a phonon. The Raman gain coefficient $g_R \approx 6 \times 10^{-14}$ m/W at $\lambda_p = 1450$ nm.

Raman Gain

$$G_R = \exp\left(\frac{g_R P_p L_{eff}}{A_{eff}}\right)$$

where P_p is pump power, $L_{eff} = [1 - e^{-\alpha_p L}]/\alpha_p$ is the effective length, A_{eff} is effective mode area.

5.2 Distributed vs Lumped Raman Amplification

- **Distributed Raman:** The transmission fiber is used as the gain medium with backward pumping. The signal is amplified continuously along the fiber, reducing the noise figure dramatically. Best OSNR performance.

- **Lumped (discrete) Raman:** A separate short high-nonlinearity fiber (HNF) provides concentrated gain. Simpler pump management.

5.3 EDFA vs Raman Amplifier Comparison

Parameter	EDFA	Raman Amplifier
Gain medium	Er-doped fiber	Transmission fiber (silica)
Gain band	1530–1565 nm (C-band)	Flexible (~ 100 nm from pump)
Pump wavelength	980/1480 nm	~ 100 nm below signal
Gain flatness	Requires GFF	Inherently broad but uneven
Noise figure	~ 3 – 6 dB	< 0 dB (effective; distributed)
Amplification type	Lumped	Distributed or lumped
Cost/complexity	Moderate	Higher pump power needed

Quick Revision — Module 4

Optical Switches, Couplers, and Amplifiers | PE-EC801B

Key Formulas at a Glance

Coupler power transfer	$P_1 = P_0 \cos^2(\kappa L); \quad P_2 = P_0 \sin^2(\kappa L)$
Coupling length	$L_c = \pi/(2\kappa)$ (full transfer)
MZM transfer function	$P_{out} = P_{in} \cos^2(\pi V/2V_\pi)$
EDFA ASE noise PSD	$S_{ASE} = n_{sp} h\nu (G - 1)$
EDFA noise figure	$NF \approx 2n_{sp}$ (large G); min = 3 dB
Raman gain	$G_R = \exp(g_R P_p L_{eff}/A_{eff})$
Raman Stokes shift	$\Delta\nu \approx 13$ THz; $\Delta\lambda \approx 100$ nm at 1550 nm

One-Line Definitions

- **Directional coupler:** 4-port device transferring power via evanescent field overlap between two waveguides.
- **Coupling coefficient κ :** Rate of power transfer per unit length in a directional coupler.
- **MZM:** Mach-Zehnder Modulator; electro-optic intensity modulator using LiNbO_3 Pockels effect.
- V_π : Voltage causing π phase shift in MZM; half-wave voltage.
- **EDFA:** Er-doped fiber amplifier; all-optical gain in 1530–1565 nm C-band.
- **ASE:** Amplified Spontaneous Emission; broadband noise generated in optical amplifiers.
- **SRS:** Stimulated Raman Scattering; pump photon excites molecular vibration, emits Stokes-shifted signal photon.
- **Distributed Raman:** Uses the transmission fiber as gain medium; best noise performance.
- **Noise figure (NF):** Degradation of OSNR through an amplifier; minimum theoretical = 3 dB.

Frequently Asked / Viva Questions

Q1. What is the principle of operation of a directional coupler?

When two waveguides are placed sufficiently close, their evanescent fields overlap. The resulting coupled mode equations show that power oscillates sinusoidally between the two guides with period $L_c = \pi/(2\kappa)$.

Frequently Asked / Viva Questions (contd.)

Q2. What is the coupling length and how is it related to κ ?

$L_c = \pi/(2\kappa)$ is the length at which all power transfers from guide 1 to guide 2. κ (m^{-1}) depends on the overlap integral of the evanescent fields.

Q3. What is the half-wave voltage V_π of an MZM?

V_π is the voltage needed to cause a phase shift of exactly π in one arm of the MZM, switching the output from maximum to minimum (or vice versa). Lower V_π means less drive power needed.

Q4. Why is LiNbO_3 used in MZMs?

LiNbO_3 has a large electro-optic (Pockels) coefficient $r_{33} \approx 30 \text{ pm/V}$, enabling efficient phase shifting with moderate voltage. It also has low optical loss in the 1310–1550 nm bands.

Q5. What are the two pump wavelengths for EDFA and their trade-offs?

980 nm: lower noise figure ($\sim 3\text{--}4 \text{ dB}$), three-level pumping, lower output power.
1480 nm: higher saturation power, two-level pumping, slightly higher NF ($\sim 4\text{--}5 \text{ dB}$). 980 nm preferred for low-noise preamplification.

Q6. What is ASE in EDFA? How does it affect system performance?

ASE is spontaneous emission from excited Er^{3+} ions that is guided and amplified. It degrades OSNR (optical signal-to-noise ratio) and limits the number of amplifier cascades in a chain.

Q7. What is the minimum theoretical noise figure of an EDFA?

3 dB. This corresponds to $n_{sp} = 1$ (complete population inversion) and large gain $G \gg 1$.

Q8. Explain stimulated Raman scattering in the context of fiber amplification.

A high-power pump photon (ν_p) interacts with a silica phonon, losing energy to produce a Stokes signal photon at $\nu_s = \nu_p - \nu_{\text{phonon}}$. If a signal is present at ν_s , it triggers more SRS events — stimulated Raman amplification.

Q9. What are the advantages of distributed Raman amplification?

The gain is spread over the entire span, keeping the signal power higher throughout the fiber. This reduces the effective noise figure and improves OSNR compared to lumped amplification. No discrete amplifier components needed in the span.

Q10. Classify optical switches and state their switching speeds.

Mechanical: ms–s; Electro-optic: ps–ns; Acousto-optic: μs ; Thermo-optic: ms; MEMS: ms.

Q11. What is the extinction ratio of an MZM?

$\text{ER} = P_{\text{max}}/P_{\text{min}}$. An ideal MZM has infinite ER (perfect extinction). Practical devices achieve 20–30 dB due to waveguide asymmetries and electrode misalignment.

EDFA vs Raman — Quick Comparison

Feature	EDFA	Raman
Gain band	1530–1565 nm	Tunable with pump
Noise figure	≥ 3 dB	< 0 dB effective (dist.)
Amplifier type	Lumped	Distributed or lumped
Gain medium	Er-doped fiber	Transmission fiber
Pump λ	980/1480 nm	~ 1450 nm (for 1550)

Module 5

WDM Fundamentals

1. WDM Fundamentals

Wavelength Division Multiplexing (WDM)

WDM is a technology that multiplexes multiple optical carrier signals on a single fiber using different wavelengths (colors) of laser light. Each wavelength carries a separate data channel, dramatically multiplying the fiber capacity without laying additional fiber.

1.1 WDM Principle and ITU-T Channel Grid

Multiple lasers operating at distinct wavelengths are combined (multiplexed) by a MUX onto a single fiber. At the far end, a DEMUX separates them back to individual receivers. Each wavelength channel is independent and carries its own data stream.

The **ITU-T G.694.1** standard defines the DWDM channel grid centered on 193.1 THz (1552.52 nm) with channel spacings of 12.5, 25, 50, or 100 GHz.

1.2 CWDM vs DWDM

Parameter	CWDM	DWDM
Full name	Coarse WDM	Dense WDM
Channel spacing	20 nm (ITU-T G.694.2)	0.4–0.8 nm (50–100 GHz)
Channel count	4–18	40–160+
Wavelength range	1271–1611 nm	1530–1565 nm (C-band)
Laser requirement	Uncooled (no TEC)	Temperature-controlled (TEC)
Cost	Low	High
Amplification	No (too wide for EDFA)	EDFA compatible
Application	Metro, short-haul	Long-haul, backbone

2. WDM Components

WDM MUX/DEMUX Devices

WDM MUX combines multiple wavelengths onto one fiber; DEMUX separates them. Key technologies: Arrayed Waveguide Grating (AWG), Thin-Film Filter (TFF), and Fiber Bragg Grating (FBG).

2.1 Arrayed Waveguide Grating (AWG)

An AWG is a planar integrated device with a star coupler at input, an array of waveguides of incrementally increasing lengths, and a second star coupler (phased array region). The path-length difference between adjacent waveguides is exactly one wavelength, causing different wavelengths to focus to different output ports due to constructive interference. AWGs are used for both MUX and DEMUX.

Advantages: Mass-producible on a single chip; very flat passband; bidirectional; scalable to 40+ channels. **Disadvantage:** Temperature-sensitive (requires TEC for DWDM).

2.2 Thin-Film Filter (TFF)

Uses multiple thin-film interference layers (Fabry-Perot cavity) to reflect one wavelength while transmitting all others. Stacked in series, each filter drops one channel. **Advantage:** Low insertion loss for the reflected channel, temperature stable. **Disadvantage:** Insertion loss accumulates for large channel counts.

2.3 Fiber Bragg Grating (FBG)

A periodic refractive index modulation written into the fiber core. Reflects only the Bragg wavelength $\lambda_B = 2n_{eff}\Lambda$ (Λ = grating period) and transmits all others. Used with circulators for wavelength drop/add.

FBG Bragg Condition

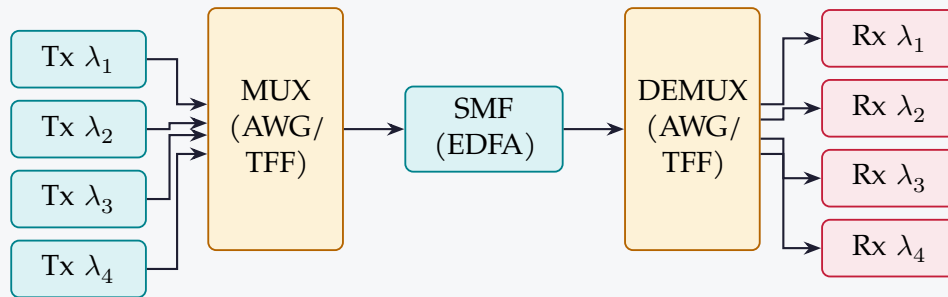
$$\lambda_B = 2n_{eff}\Lambda$$

where n_{eff} is the effective refractive index of the guided mode and Λ is the grating period. Tunable by temperature or strain.

2.4 Optical Add-Drop Multiplexer (OADM)

An OADM allows one or more wavelength channels to be dropped from and added to a WDM signal at an intermediate node without converting to electrical signals. A **ROADM** (Reconfigurable OADM) allows software-controlled selection of any channel.

WDM System Block Diagram — 4-Channel DWDM



3. WDM Network Principles — OXC and Protection

Optical Cross-Connect (OXC)

An **OXC** is a network node that switches wavelength channels from any input fiber to any output fiber, entirely in the optical domain. It is the building block of all-optical mesh networks and enables wavelength routing.

Mesh topology and protection:

- **Mesh topology:** Each node connects to multiple other nodes via multiple fibers. Provides multiple routing paths for each wavelength channel.
- **Protection switching:** Dedicated 1+1 protection pre-assigns a backup path. On failure, traffic is immediately switched to backup (<50 ms).
- **Restoration:** Dynamic rerouting of traffic around a failure using spare capacity. Takes longer (>100 ms) but uses network resources more efficiently.

4. Nonlinear Effects in Optical Fiber

Nonlinear Effects in Fiber

At high optical power, the refractive index of silica becomes intensity-dependent: $n = n_0 + n_2 I$, where $n_2 \approx 2.6 \times 10^{-20} \text{ m}^2/\text{W}$ is the nonlinear refractive index coefficient. This gives rise to three key nonlinear phenomena: SPM, XPM, and FWM.

4.1 Self-Phase Modulation (SPM)

A pulse modulates its own phase due to the intensity-dependent refractive index. The nonlinear phase shift accumulated is:

$$\phi_{NL} = \gamma P_0 L_{eff}, \quad \gamma = \frac{n_2 \omega}{c A_{eff}} = \frac{2\pi n_2}{\lambda A_{eff}}$$

where γ is the fiber nonlinearity coefficient, P_0 is peak power, L_{eff} is the effective length.

B-integral (total accumulated nonlinear phase):

$$B = \gamma P_0 L_{eff}$$

SPM causes **spectral broadening** of the pulse. The leading edge of the pulse is red-shifted; the trailing edge is blue-shifted. In the anomalous dispersion regime, SPM and dispersion can balance to form solitons.

4.2 Cross-Phase Modulation (XPM)

In a WDM system, the intensity of one channel modulates the phase of an adjacent channel. The nonlinear phase induced in channel 2 by channel 1 is $2\gamma P_1 L_{eff}$ (factor of 2 due to polarization mixing). XPM causes amplitude-to-phase conversion, broadens spectra, and creates timing jitter in WDM channels.

4.3 Four-Wave Mixing (FWM)

When three channels at frequencies $\omega_1, \omega_2, \omega_3$ co-propagate, a new frequency $\omega_4 = \omega_1 + \omega_2 - \omega_3$ is generated (and other combinations). This creates crosstalk between WDM channels.

FWM Phase Matching Condition

FWM is most efficient when the **phase mismatch** $\Delta\beta$ is small:

$$\Delta\beta = \beta(\omega_1) + \beta(\omega_2) - \beta(\omega_3) - \beta(\omega_4) \approx 0$$

In terms of the dispersion parameter D :

$$\Delta\beta \approx \frac{2\pi c D}{\lambda^2} \Delta\lambda^2 \cdot \lambda^2$$

FWM is strongest near the zero-dispersion wavelength (small D) and for equal channel spacing. Mitigation: use non-zero dispersion-shifted fiber (NZ-DSF) with $D \neq 0$ in the signal band.

5. Group Velocity Dispersion (GVD) and Dispersion Parameter

GVD and Dispersion Parameter D

Group velocity dispersion (GVD) refers to the variation of group velocity with frequency, parameterized by $\beta_2 = d^2\beta/d\omega^2$ (ps²/km). The **dispersion parameter** D relates directly to β_2 : $D = -\frac{2\pi c}{\lambda^2} \beta_2$ (ps/nm·km).

- **Normal dispersion:** $\beta_2 > 0$ ($D < 0$); shorter wavelengths travel faster; pulse broadens normally.
- **Anomalous dispersion:** $\beta_2 < 0$ ($D > 0$); longer wavelengths travel faster; enables soliton formation.
- **Zero-dispersion wavelength λ_0 :** For standard SMF, $\lambda_0 \approx 1310$ nm. In DSF, shifted to 1550 nm.

Dispersion Length

$$L_D = \frac{T_0^2}{|\beta_2|}$$

where T_0 is the initial pulse half-width (1/e point). L_D is the length at which the pulse broadens by $\sqrt{2}$. For $L \ll L_D$, dispersion is negligible.

6. Soliton-Based Communication**Optical Soliton**

An **optical soliton** is a pulse that propagates through a dispersive fiber without changing its shape, because the effects of anomalous GVD (pulse spreading) are exactly balanced by SPM (pulse compression). Solitons are solutions to the Nonlinear Schrödinger Equation (NLSE).

6.1 Nonlinear Schrödinger Equation (NLSE)

The propagation of an optical pulse in a fiber is governed by:

$$j\frac{\partial A}{\partial z} - \frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + \gamma|A|^2A = 0$$

where $A(z, T)$ is the slowly-varying pulse envelope, $\beta_2 < 0$ for anomalous dispersion, and γ is the nonlinear coefficient.

6.2 Fundamental Soliton (N = 1)

The fundamental soliton has the **sech** profile:

$$A(0, T) = \sqrt{P_0} \operatorname{sech}(T/T_0)$$

The soliton condition relates dispersion and nonlinear lengths:

Soliton Condition and Width-Power Relation

$$L_D = L_{NL}, \quad \text{i.e.,} \quad \frac{T_0^2}{|\beta_2|} = \frac{1}{\gamma P_0}$$

$$\Rightarrow P_0 = \frac{|\beta_2|}{\gamma T_0^2}$$

The peak power required for a soliton increases as the pulse width T_0 decreases. For $T_0 = 10$ ps, $|\beta_2| = 20$ ps²/km, $\gamma = 2$ W⁻¹km⁻¹: $P_0 = 20/(2 \times 100) = 0.1$ W.

Higher-order solitons ($N = 2, 3, \dots$) periodically breathe (compress and expand) with soliton period $z_s = \pi L_D/2$.

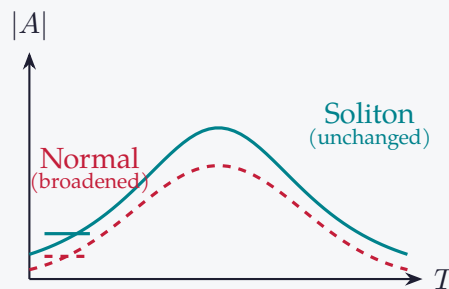
6.3 Advantages and Limitations of Soliton Communication

Advantages:

- Dispersion-free propagation: pulse shape preserved over trans-oceanic distances.
- Allows very high bit rates (sub-picosecond pulses).
- With periodic amplification, solitons can be maintained over global distances.

Limitations:

- Requires anomalous dispersion ($D > 0$) throughout the link; problematic with DCF.
- Gordon-Haus jitter: ASE noise from amplifiers causes random timing jitter, limiting transmission distance.
- Soliton-soliton interaction: adjacent pulses at the same wavelength must be spaced $> 5T_0$ apart.
- Requires precise control of launch power and pulse shape.
- Dispersion management (DM-solitons) is needed for WDM compatibility.

Soliton Propagation vs Normal Pulse Propagation

Quick Revision — Module 5

WDM Systems and Nonlinear Effects | PE-EC801B

Key Formulas at a Glance

FBG Bragg condition	$\lambda_B = 2n_{eff}\Lambda$
SPM nonlinear phase	$\phi_{NL} = \gamma P_0 L_{eff}$ (B-integral)
Fiber nonlinearity coeff.	$\gamma = 2\pi n_2 / (\lambda A_{eff})$ ($\text{W}^{-1}\text{km}^{-1}$)
Dispersion parameter	$D = -(2\pi c / \lambda^2) \beta_2$ (ps/nm·km)
Dispersion length	$L_D = T_0^2 / \beta_2 $
Nonlinear length	$L_{NL} = 1 / (\gamma P_0)$
Soliton condition	$L_D = L_{NL}$
Soliton peak power	$P_0 = \beta_2 / (\gamma T_0^2)$
Soliton shape	$A(0, T) = \sqrt{P_0} \text{sech}(T/T_0)$
DWDM channel spacing	50–100 GHz (ITU-T G.694.1)

One-Line Definitions

- **WDM:** Multiple wavelength channels on a single fiber; multiplies capacity without new fiber.
- **DWDM:** Dense WDM with 50/100 GHz channel spacing; supports 40–160+ channels in C-band.
- **AWG:** Arrayed Waveguide Grating; planar integrated MUX/DEMUX using path-length differences.
- **FBG:** Fiber Bragg Grating; periodic index modulation reflecting only the Bragg wavelength.
- **OADM:** Optical Add-Drop Multiplexer; drops and adds wavelength channels at intermediate nodes.
- **OXC:** Optical Cross-Connect; switches wavelengths between fibers in an all-optical network node.
- **SPM:** Self-Phase Modulation; pulse modulates its own phase, causing spectral broadening.
- **XPM:** Cross-Phase Modulation; one WDM channel modulates the phase of another.
- **FWM:** Four-Wave Mixing; three channels generate a fourth frequency; crosstalk near λ_0 .

- **Soliton:** Pulse that propagates undistorted due to exact SPM–GVD balance (anomalous dispersion).
- β_2 : GVD parameter; $\beta_2 < 0$ = anomalous; $\beta_2 > 0$ = normal dispersion.

Frequently Asked / Viva Questions

Q1. What is WDM and what are its advantages?

WDM multiplexes multiple wavelength channels on one fiber. Advantages: multiplies capacity N -fold without laying new fiber; each channel is protocol- and bit-rate-independent; can combine CWDM and DWDM in hybrid networks.

Q2. Differentiate between CWDM and DWDM.

CWDM: 20 nm channel spacing, 4–18 channels, 1271–1611 nm, uncooled lasers, low cost, no EDFA. DWDM: 50–100 GHz spacing, 40–160+ channels, 1530–1565 nm, TEC-cooled lasers, EDFA-compatible, high capacity, high cost.

Q3. What is an AWG and how does it work?

An Arrayed Waveguide Grating is a planar integrated optics device. Light enters a free propagation region (star coupler), splits into an array of waveguides with linearly increasing lengths, recombines in a second star coupler. The path-length differences cause different wavelengths to constructively interfere at different output ports.

Q4. What is an FBG? State the Bragg condition.

A Fiber Bragg Grating is a periodic modulation of the refractive index written into the fiber core using UV light. It reflects only $\lambda_B = 2n_{eff}\Lambda$ and transmits all others. Used for channel dropping in OADMs, dispersion compensation, and sensing.

Q5. What is Self-Phase Modulation? What does it cause?

SPM arises because $n = n_0 + n_2 I$. A pulse modulates its own phase: $\phi_{NL}(t) = \gamma P(t)L_{eff}$. The leading edge is red-shifted and trailing edge is blue-shifted, causing spectral broadening. In anomalous dispersion, SPM enables soliton formation.

Q6. What is Four-Wave Mixing and when is it worst?

FWM generates new frequencies $\omega_4 = \omega_1 + \omega_2 - \omega_3$ from three co-propagating channels. It is worst when the phase mismatch $\Delta\beta \approx 0$, i.e., near the zero-dispersion wavelength λ_0 . Equal channel spacing also maximizes FWM efficiency. NZ-DSF ($D \neq 0$) mitigates it.

Q7. Define the dispersion parameter D and state its units.

$D = -(2\pi c/\lambda^2)\beta_2$ in units of ps/nm·km. It gives the pulse broadening per nm of source linewidth per km of fiber. Standard SMF: $D \approx +17$ ps/nm·km at 1550 nm.

Q8. What is anomalous dispersion and why is it needed for solitons?

Anomalous dispersion ($\beta_2 < 0$, $D > 0$): longer wavelengths travel faster. The chirp induced by anomalous GVD (red leading) is opposite to SPM chirp

Frequently Asked / Viva Questions (contd.)

(red leading). At the right power, these two effects exactly cancel, producing a shape-invariant soliton.

Q9. State the soliton condition and derive the soliton power.

Soliton condition: $L_D = L_{NL}$, i.e., $T_0^2/|\beta_2| = 1/(\gamma P_0)$. Solving: $P_0 = |\beta_2|/(\gamma T_0^2)$. For $T_0 = 10$ ps, $|\beta_2| = 20$ ps²/km, $\gamma = 2$ W⁻¹km⁻¹: $P_0 = 0.1$ W.

Q10. What are the limitations of soliton communication?

Gordon-Haus timing jitter (from ASE noise), soliton-soliton interaction requiring large pulse spacing ($> 5T_0$), requirement for precisely anomalous dispersion throughout, and incompatibility with standard DWDM channels without dispersion-managed fiber.

Q11. What is XPM and how does it differ from SPM?

XPM: the intensity of channel i modifies the phase of channel j with a coefficient of 2γ (vs γ for SPM). XPM causes amplitude-dependent phase noise and timing jitter in WDM systems, particularly at high channel counts.

Q12. What is an OADM and what is its purpose in WDM networks?

An Optical Add-Drop Multiplexer allows one or more wavelength channels to be dropped to and added from a local node while the remaining channels pass through transparently. A ROADM adds software reconfigurability, supporting any-to-any wavelength routing.

Nonlinear Effects — Summary Comparison

Effect	Mechanism	Impact	Mitigation
SPM	$n_2 I$ on own pulse	Spectral broadening	Reduce power; use DSF
XPM	$n_2 I$ from other channel	Phase noise, jitter	Channel spacing, filtering
FWM	3-wave mixing $\rightarrow$ 4th	Crosstalk; power drain	NZ-DSF; unequal spacing
SRS	Raman scattering	Power transfer to lower λ	Limit span power